
AI-Powered Document Insights and Data Extraction

PharmaDoc Intelligence

Source-grounded Q&A for digital and scanned pharmaceutical PDFs

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OPEN-SOURCE DOCUMENT Q&A

PharmaDoc Intelligence

Source-grounded answers from digital and scanned pharmaceutical documents

Document Workspace

Upload one or more PDFs



Sample Documentation.pdf

6.9 MB ↓

Process Documents

Clear Documents

Files processed: 1

Total pages: 39

OCR pages: 39

Chunks indexed: 50

Processing time: 280.29 seconds

Document types: BSE/TSE Declaration: 1, Certificate of Quality:

2, Cover Letter: 1, Other: 1

PDF preview

Analysis mode

Ask a Question

Advanced retrieval

Document Q&A

Chatbot

Process one or more PDFs, then ask a question about their contents.

Question or optional focus

Enter a question, or leave blank when using Key Findings or Summarise Document...

Ready for a question

Ask Assistant

Expand Chat

Clear Chat

PharmaDoc Intelligence
Interactive pharmaceutical document Q&A interface

Problem Statement | Pharmaceutical Document Intelligence

Extracting reliable, source-grounded information from varied pharmaceutical PDFs.

Challenge

- Mixed digital and scanned PDFs
- Tables and inconsistent layouts
- Rotated pages and mixed document types
- Difficulties extracting key regulated information
- OCR and layout issues can affect reliability

Solution

- Direct PDF extraction + orientation-aware OCR
- Chunking + MiniLM embeddings
- FAISS retrieval + document routing
- Local Phi-3.5 answer generation
- Page-level sources + retrieval evidence

Pain Point

Scanned / image PDFs

Rotated pages

Long mixed documents

Exact regulated fields

How the Pipeline Solves It

Tesseract OCR fallback

Orientation-aware OCR

Chunking + document routing

Targeted field extraction + sources

System Architecture | OCR-Enhanced Pharmaceutical RAG Pipeline



Component Specifications

Component	Technology	Configuration
OCR	Tesseract	English; PSM 3; orientation detection for rotated scans
Chunking	LlamaIndex SentenceSplitter	500-character chunks; 100-character overlap
Embeddings	all-MiniLM-L6-v2	Normalised dense embeddings
Vector index	FAISS IndexFlatIP	In-memory inner-product similarity
Retriever	Metadata-aware dense retrieval	Top-K = 4 by default; document-type routing
LLM	Microsoft Phi-3.5-mini-instruct	Local GPU inference; deterministic; max 400 new tokens
Prompting	Grounded RAG prompt	Retrieved context + page-level sources + exact-field evidence

Pipeline Performance Metrics | Final Targeted Test Set

Manual evaluation of the final tested queries and observed system timings.

Retrieval Performance

- Hit rate: 100%
- Retrieval confidence range: 58%–82% across targeted tests
- Top-K retrieval: up to 4 chunks per query
- Relevant source evidence retrieved for all targeted tests

End-to-End Accuracy

- Answer accuracy: 100% (4/4 targeted tests)
- Citation accuracy: 100% in targeted tests
- Broad free-form summary exposed one known interpretation limitation

System Performance

- Targeted answer times: 2.41–6.59 s
- Average targeted answer time: 4.31 s
- 39-page OCR processing: 280.29 s
- Broad summary example: 16.88 s

Tested examples: Certificate dates and BSE/TSE compliance from Sample Documentation.pdf; Process Run ID from pharmaceutical-sdf-page6-dosimetry-rotated.pdf; and delivered-dose extraction from pharmaceutical-sdf-page4-certificate-processing.pdf.

Pipeline in Action | Factual Lookup

Query

“What is the manufacturing date, release date, and retest date in the Certificate of Quality?”

Retrieved evidence:

4 chunks from Certificate of Quality pages 4, 5, 4, and 7

Retrieval confidence: 82% (High)

Final answer:

Manufacturing: 06/17/2025

Release: 08/29/2025

Retest: 06/16/2029

Correct multi-field extraction with page-level source attribution in 6.59 seconds.

The screenshot shows a chatbot window with a title bar 'Chatbot' and a close button. The user's query is 'What is the manufacturing date, release date, and retest date in the Certificate of Quality?'. The chatbot's response is as follows:

The manufacturing date is 06/17/2025, the release date is 08/29/2025, and the retest date is 06/16/2029. These dates pertain to the Certificate of Quality issued by Basf Corporation.

Sources

- [Source 1] Sample Documentation.pdf – Certificate of Quality, Page 4
- [Source 2] Sample Documentation.pdf – Certificate of Quality, Page 5
- [Source 3] Sample Documentation.pdf – Certificate of Quality, Page 4
- [Source 4] Sample Documentation.pdf – Certificate of Quality, Page 7

Retrieval details

Retrieval confidence: 82% (High)
Evidence coverage: 4 of 7 available chunks
Chunks used: 4
Route: Certificate of Quality (Keyword)
Response time: 6.59 seconds

At the bottom of the chatbot window, there is a small icon of a document and a downward arrow.

Pipeline in Action | Compliance / Multi-Part Query

Query

“Does Kollidon SR contain any material of animal or human origin, and which BSE/TSE guidance does it meet?”

Retrieved context:

4 BSE/TSE Declaration chunks from pages 10, 23, 39, and 14
Retrieval confidence: 72% (High)

Response quality:

Correctly states that Kollidon SR contains no animal/human-origin material and identifies EMEA/410/01.

Complete two-part compliance answer with cited evidence in 5.23 seconds.

The screenshot displays a chatbot interface with a 'retrieval details' section at the top. Below this, a user query is shown in a light blue bubble. The chatbot's response follows, including a paragraph of text and a list of four sources. A second 'retrieval details' section is at the bottom.

retrieval details

Retrieval confidence: 82% (High)
Evidence coverage: 4 of 7 available chunks
Chunks used: 4
Route: Certificate of Quality (Keyword)
Response time: 6.59 seconds

Does Kollidon SR contain any material of animal or human origin, and which BSE/TSE guidance does it meet?

Kollidon SR does not contain any material of animal or human origin. It meets the requirements of "EMA/410/01" regarding the risk of transmitting animal spongiform encephalopathy agents via human and veterinary medicinal products.

Sources

- [Source 1] Sample Documentation.pdf – BSE/TSE Declaration, Page 10
- [Source 2] Sample Documentation.pdf – BSE/TSE Declaration, Page 23
- [Source 3] Sample Documentation.pdf – BSE/TSE Declaration, Page 39
- [Source 4] Sample Documentation.pdf – BSE/TSE Declaration, Page 14

Retrieval details

Retrieval confidence: 72% (High)
Evidence coverage: 4 of 36 available chunks
Chunks used: 4
Route: BSE/TSE Declaration (Keyword)
Response time: 5.23 seconds

Design Decision Analysis | Reliable Answers on Limited Compute

Model Selection Reasoning

Decision	Rationale	Trade-off
MiniLM embeddings	Fast, lightweight semantic retrieval	Less expressive than larger embeddings
500 / 100 chunking	Balances context and retrieval granularity	Can split complex tables
Phi-3.5 mini	Open-source and practical on Colab GPU	Less capable than larger hosted LLMs
FAISS in memory	Simple, fast, no external service	Not persistent / distributed at scale

Speed vs. Accuracy

- Used compact local models to keep the pipeline practical in Colab.
- Added exact-field extraction where precision mattered more than free-form generation.

Complexity vs. Maintainability

- Kept FAISS and fixed-size chunking simple and transparent.
- Added document routing and orientation-aware OCR only after testing showed clear value.

Current Limitations & Next Steps | Evidence-Led Improvement Roadmap

Current Limitations

- Dense scanned tables can still be misinterpreted during broad summarisation, even when OCR and retrieval succeed.
- OCR quality remains sensitive to scan clarity; orientation handling fixed the rotated-page failure, but very sparse or degraded scans may remain difficult.
- Document taxonomy is coarse: a Certificate of Processing can route as Certificate of Quality.
- The local LLM can be slower on open-ended synthesis than on targeted field questions.

Proposed Enhancements

- | | |
|--------------------|--|
| Short term | Add table/layout-aware extraction and stronger summary validation. |
| Medium term | Evaluate reranking and alternative embeddings; separate retrieval from generation latency. |
| Long term | Persistent vector storage, larger evaluation set, monitoring, and integration into regulated document workflows. |

Testing-driven fix implemented

Rotated dosimetry OCR → orientation-aware OCR → correct
Process Run ID 129708B.

Project Impact & Learning Outcomes

Technical Learning

- Built an end-to-end RAG workflow in Python.
- Applied OCR, embeddings, FAISS retrieval, prompt engineering, and a local LLM.
- Diagnosed a real rotated-OCR failure and implemented a general fix.

Potential Business Impact

- Faster access to manufacturing, quality, and compliance facts across long PDFs.
- Source-grounded answers reduce manual searching and make evidence easier to verify.
- Could support document review while keeping a human in the loop for regulated decisions.

Skills Developed

- Python and Google Colab
- RAG and semantic retrieval
- OCR and document intelligence
- Gradio UI development
- Testing, debugging, and evaluation
- Problem-solving and technical documentation

From unstructured pharmaceutical PDFs to traceable answers — improved through testing.

Thank You!

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